Dr. Hassan Hijry

PH.D. in Systems Engineering

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PROFILE

HASSAN HIJRY is an Assistant Professor of Industrial Engineering at the University of Tabuk, Saudi Arabia, with an MS in Industrial Engineering from Lawrence Technological University (LTU), and a Ph.D. in Systems Engineering from Oakland University, USA. He brings industry experience as a former frontline manager at PEPSICO in Al-Riyadh. His teaching expertise spans Work Study, Production Planning and Control, Facilities Planning, Materials Handling, and Industry 4.0 Technologies and Engineering Management. Hassan's research focuses on Industry 4.0, AI, process optimization, systems analysis, and data-driven decision-making across diverse workplaces, manufacturing industries, and healthcare systems.

ACADEMIC EXPERIENCE:

Teacher Assistant, University of Tabuk — May, 2013 — Nov 2013

Research Assistant – HCI Lab, Oakland University — Aug, 2018 — Dec 2018

Teacher Assistant **-** Oakland University — May, 2019 — Aug 2019

Research Assistant - SEAL Lab , Oakland University — Jan, 2020 — May 2020

Assistant Professor, University of Tabuk — Aug, 2022 - present

INDUSTRIAL EXPERIENCE

Front Line Manager, PepsiCo — **May 2012 — Dec 2012**

Al-Riyadh, SA

EDUCATION

Oakland University — **Aug 2022**
Ph.D. in Systems Engineering, MI, USA

Lawrence Technological University — **May 2017**
Master's Degree, MI, USA

FUNDED PROJECTS

1. 0276-2024-S: AI-Prognostics and Health Management for Predictive Maintenance: Enhancing Efficiency and Sustainability of Vital Industrial Systems with Advanced Decision-Making Tools.

AWARDS & HONORS

2023
Recognition and Judging Panel Member, Ultimate Innovation Program (Second Cohort), University of Tabuk.

2024
Recognition Award, Deanship of E-Learning, University of Tabuk, for supervising the Community Learning Project (Academic Year 1445H / 2023–2024).

2024
Outstanding Teaching Award in Human Factors and Ergonomics, 14th IEOM International Conference, Dubai, UAE.

2025
Research Excellence Recognition (publishing in high-quality journals), University of Tabuk, for multiple awarded research projects in smart manufacturing, IoUT, sustainable materials, and human factors.

2026
Distinguished Conference Leadership Award, IEOM 2026 Conference, University of Tabuk, Saudi Arabia. Recognized for outstanding leadership and contribution to the organization and the success of the conference.

ACADEMIC LEADERSHIP & SERVICE

Keynote Speaker at the 24th Hajj, Umrah, and Ziyarah Research Conference, Umm Al-Qura University, Makkah, Saudi Arabia, 2024.

Program Committee Member, International Conference on ICT Application Research (IAR 2024), Aomori, Japan, August 31 – September 3, 2024.

Co-Chair and organizing committee, 3rd GCC International Conference on Industrial Engineering and Operations Management (IEOM 2026), Tabuk, Saudi Arabia, February 2–4, 2026.

Faculty Advisor, IEOM Student Chapters in Saudi Arabia, University of Tabuk (2022–Present**)**

ONGOING RESEARCH & CONSULTANCY PROJECTS

Ground Vehicle Dynamic Modeling and Advanced Control Using Fuzzy Logic and ANFIS-Based Inverse Learning.

Crisis and Emergency Management Research Project for the Tabuk Region, in collaboration with the Emirate of Tabuk (Towards Establishing a Regional Crisis and Emergency Management Center).

AI and Machine Learning–Based Predictive Maintenance (Industry Collaborative Research Project).

MASTER'S THESIS SUPERVISION

Detection of Thermal Anomalies in Bus Transit Corridors Using Machine Learning Techniques, 2024.

Rehabilitation and Development of Urban Wadis: A Comprehensive Study on Transforming Wadis into Safe and Sustainable Recreational Spaces in Urban Environments – Wadi Daba'an (Tabuk City), 2025.

CONFERENCE PAPER PRESENTATION

Hijry, H., et al. (2026). *Comparative study of LQR and FHLQT for ground vehicle lane change maneuvers*. In **Proceedings of the 3rd GCC International Conference on Industrial Engineering and Operations Management (GCC IEOM 2026)**, Tabuk, Saudi Arabia, February 2–4, 2026. (In progress).

Hijry, H., & Elnashar, G. A. (2026). *Ground vehicle dynamic modeling and health-aware advanced control using fuzzy logic and ANFIS-based inverse learning*. In **Proceedings of the 10th IEEE Conference on Control Technology and Applications (CCTA 2026)**, August 12–14, 2026. (Under review).

Hijry, H., & Olawoyin, R. (2025). *Human-centered stress monitoring in smart assembly lines using explainable AI with SHAP*. In **Proceedings of the ASME 2025 20th International Manufacturing Science and Engineering Conference (MSEC 2025)**. https://doi.org/10.1115/MSEC2025-154225

Hijry, H. (2024). *Proposed model for NEOM city based on Internet of Things (IoT) and MLC at ED system*. In **Proceedings of the 2024 IEEE International Conference on Prognostics and Health Management (ICPHM 2024)**. DOI: 10.1109/ICPHM61352.2024.10627366

Hijry, H., & Fares, S. A. (2024). *Implementing an AI-driven ED system for efficient emergency management during Hajj*. In **Proceedings of the 24th Scientific Forum for Hajj, Umrah and Madinah Visit Research** (2024).

Alkahlani, R. K., Almaliki, K. H., Abuduhair, A. N., **Hijry, H.,** & Aljohani, M. O. (2024). *Revolutionizing last-mile logistics: A conceptual model for smart parcel delivery with contactless innovation*. In **Proceedings of the 2nd GCC International Conference on Industrial Engineering and Operations Management (GCC IEOM 2024)**. https://doi.org/10.46254/GC02.20240175

Hijry, H. (2023). *Multi-objective optimization approach for healthcare system design configuration*. In **Proceedings of the 11th Annual International Conference on Industrial, Systems and Design Engineering**, Athens, Greece, June 19–22, 2023. https://www.atiner.gr/present

Hijry, H., & Olawoyin, R. (2020). *Application of machine learning algorithms for patient length of stay prediction in emergency department during Hajj*. In **Proceedings of the 2020 IEEE International Conference on Prognostics and Health Management (ICPHM 2020)**. DOI: 10.1109/ICPHM49022.2020.9187055

CERTIFICATIONS OR PROFESSIONAL REGISTRATIONS:

- A Workshop on Assessment and Evaluation of Student Learning for Program Improvement (2024)
- AI & Supply Chain Analysis in Smart Manufacturing, OU/IBM training (2022)
- Predictive Maintenance Hands-On Workshop, OU/IBM training (2022)
- Growth Lessons from the Refactoring Research Community, Seminar, OU (2022)
- Simulation Modeling and Analysis, Workshop Program, FlexSim Software (2020)
- Realizing the Digital Enterprise Seminar, Siemens PLM (2019)
- Lean Approach to Quality Engineering Systems, LTU, U.S.A (2016)
- Occupational Safety and Health Administration -OSHA- LTU, U.S.A (2016)
- Lean Six Sigma DIMAC Methodology, Green Belt, LTU, U.S.A (2015)
- Maintenance Training Program, Saudi Arabian Airlines, Jeddah, Saudi Arabia (2012)

PRINCIPAL PUBLICATIONS OF THE LAST FIVE YEARS:

Hijry, H. (In press). Multi-class fault classification in conveyor systems using machine learning: Enhancing reliability in production logistics. *Acta Logistica*. Manuscript ID: AL-780-2025.

Hijry, H., Mohsen, S., Albalawi, O., Ali, S., Alhawiti, A., & Elnashar, G. (2026). *Development and simulation of a novel mathematical model for an intelligent tire system toward predictive maintenance*. *Scientific Reports*. https://doi.org/10.1038/s41598-026-38625-1

Hijry, H. (2025). *Advancing sustainable smart manufacturing: A comprehensive review of machine learning techniques in assembly lines*. *Sustainability, 18*(1), 348. https://doi.org/10.3390/su18010348

Alnahas, J., Lakhouit, A., Alnchiwati, M., Albalawi, M., Elrazzaz, A., Alanzi, N., Alghamdi, A., Yassin, A. H. A., & **Hijry, H.** (2025). *Enhancing the properties of concrete with the incorporation of recycled polypropylene plastic*. *Buildings, 15*(7), 1055. https://doi.org/10.3390/buildings15071055

Hijry, H., Naqvi, S. M. R., Javed, K., Albalawi, O. H., Olawoyin, R., Varnier, C., & Zerhouni, N. (2024). *Real time worker stress prediction in a smart factory assembly line*. *IEEE Access, 12*, 116238-116249. https://doi.org/10.1109/ACCESS.2024.3446875

Zayed, M. M., Mohsen, S., Alghuried, A., **Hijry, H**., & Shokair, M. (2024). *IoUT-oriented efficient CNN model for modulation schemes recognition in optical wireless communication systems*. *IEEE Access, 12*, 186836-186855. https://doi.org/10.1109/ACCESS.2024.3515895

Hijry, H., & Olawoyin, R. (2022). *Simulation model for patient workflow at selected emergency room during Hajj*. *International Journal of Simulation: Systems, Science & Technology, 22*(4). https://edas.info/doi/10.5013/IJSSST.a.22.04.08

Hijry, H., Olawoyin, R., McDonald, G., Debnath, D., Edward, W., & Al-Hejri, Y. (2021). *Predicting average wait-time of COVID-19 test results and efficacy using machine learning algorithms*. *International Journal of Industrial Engineering and Operations Management*, 75–88. https://doi.org/10.46254/j.ieom.20210202

Hijry, H., & Olawoyin, R. (2021). *Predicting patient waiting time in queuing system using multi deep learning optimization algorithms in the emergency room*. *International Journal of Industrial Engineering and Operations Management*, 33–45. https://doi.org/10.46254/j.ieom.20210103

MEMBERSHIP IN PROFESSIONAL ORGANIZATIONS:

IEOM
IEEE
IISE